

Avoiding Statistical & Methodological Mistakes in Your Research

Tim Houle, Ph.D.

~~Avoiding~~ *Finding* Statistical & Methodological Mistakes in *Published* ~~Your~~ Research

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Disclosures

- None

Topics that are commonly seen in the anesthesiology literature

- Gathered from numerous articles, and my own experience in evaluating studies
- The topics are issues related to common (frequentist) methods; only brief mention of other methods
- The 10 topics will be presented in a rank order of importance
 - Added drama
 - Prevent boredom from killing you

Poor reporting is bad and getting worse



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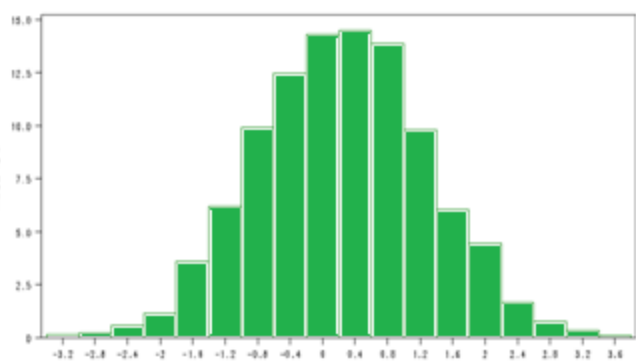
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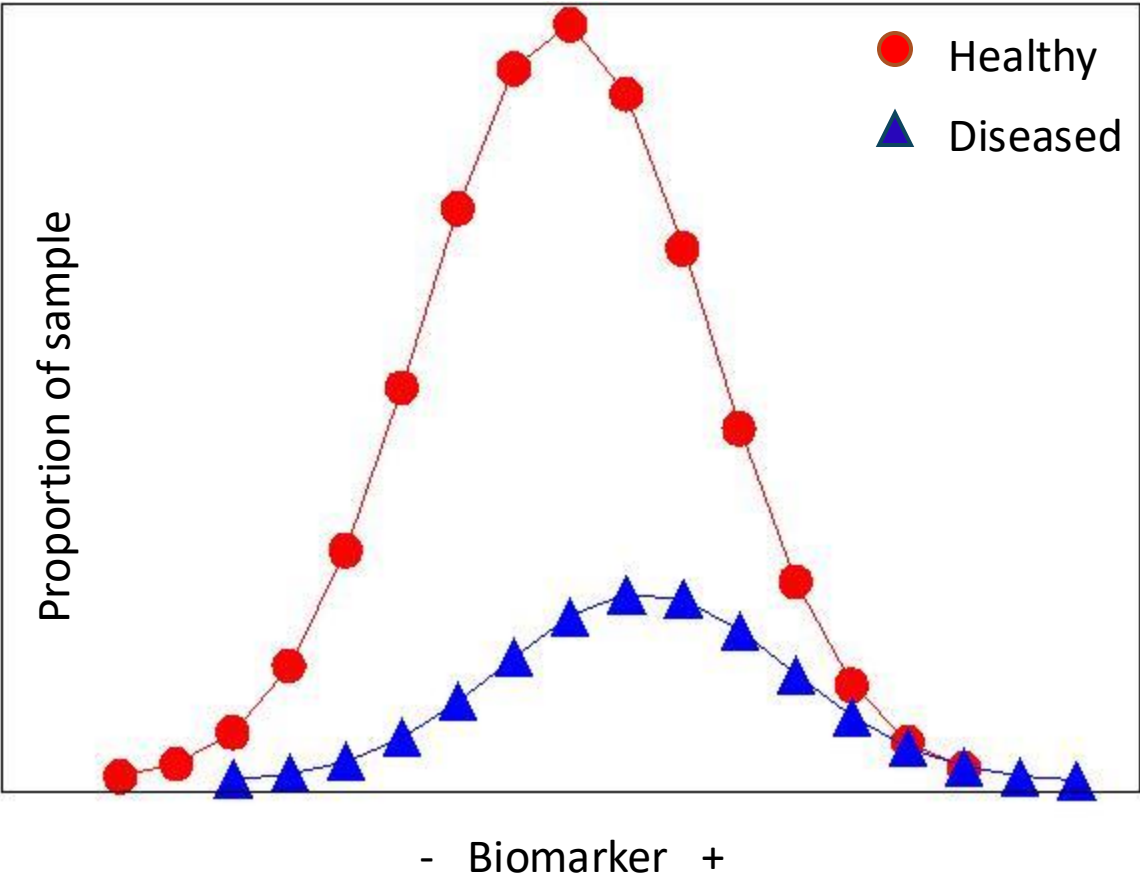
Lack of detail about the analytical plan

- Provide enough detail to allow replication if allowed access to your data:
 - Statistical Software
 - Criterion for statistical significance ($p < 0.05$)
 - Hypothesis testing (two-tailed)
 - Assumptions underlying tests (normality, etc.)
 - Actual applied tests, not just a list
 - Reporting of data (mean, median, etc.)
 - Correction for multiplicity

Lack of attention to underlying assumptions of utilized methods

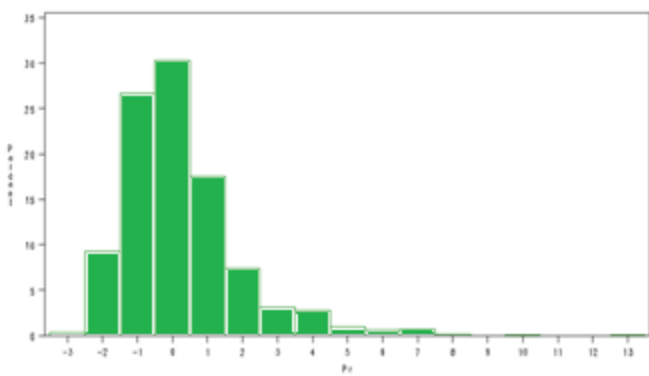


Mean_{diff} = 0.50 SD

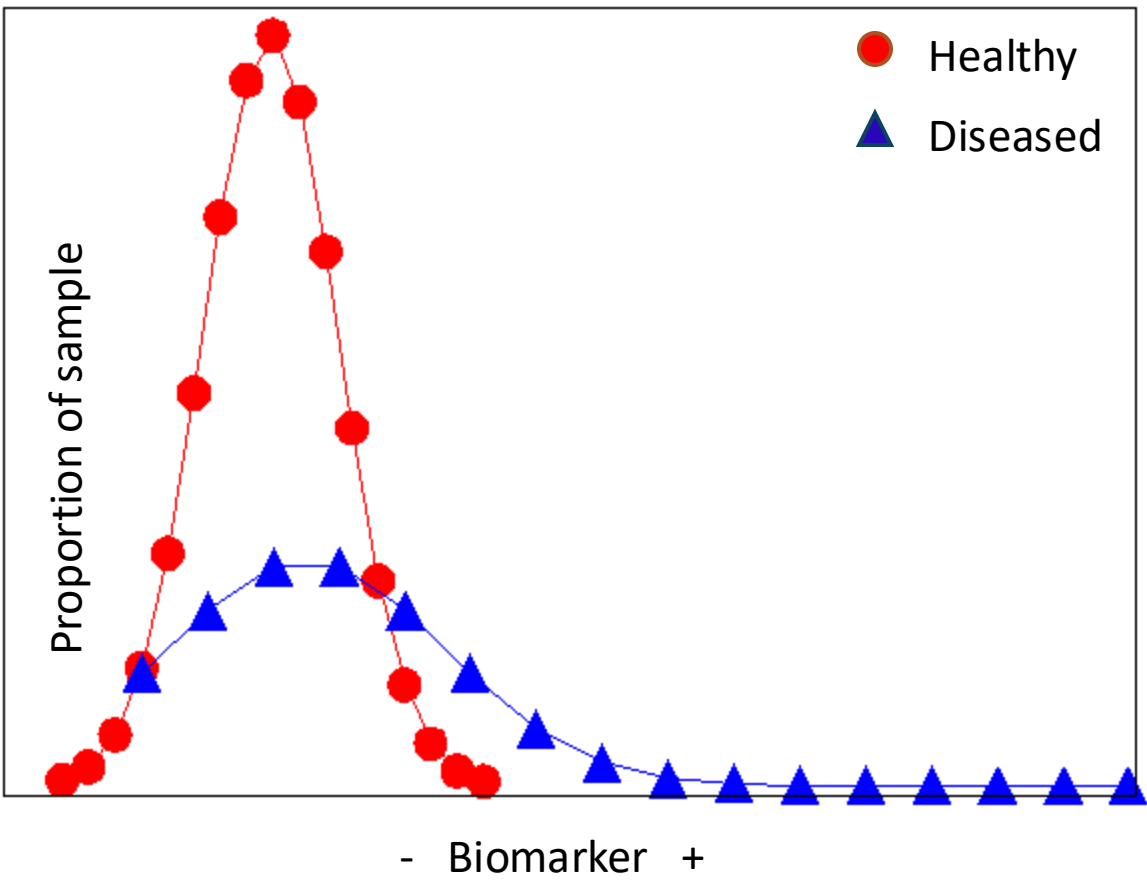


Common: Evaluate the mean difference
In a biomarker in the “Healthy” versus the “Diseased” using t-test.

The distribution of the biomarker: which statistical test/method to use for screening?



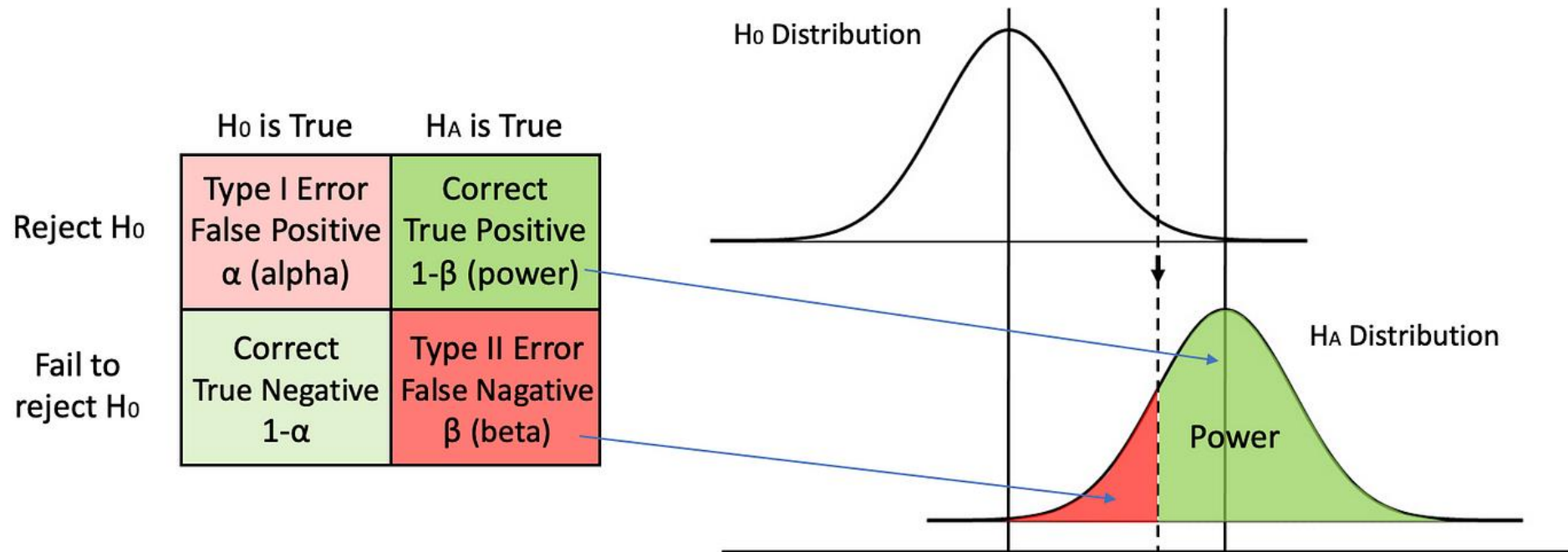
Mean_{diff} = 0.50 SD

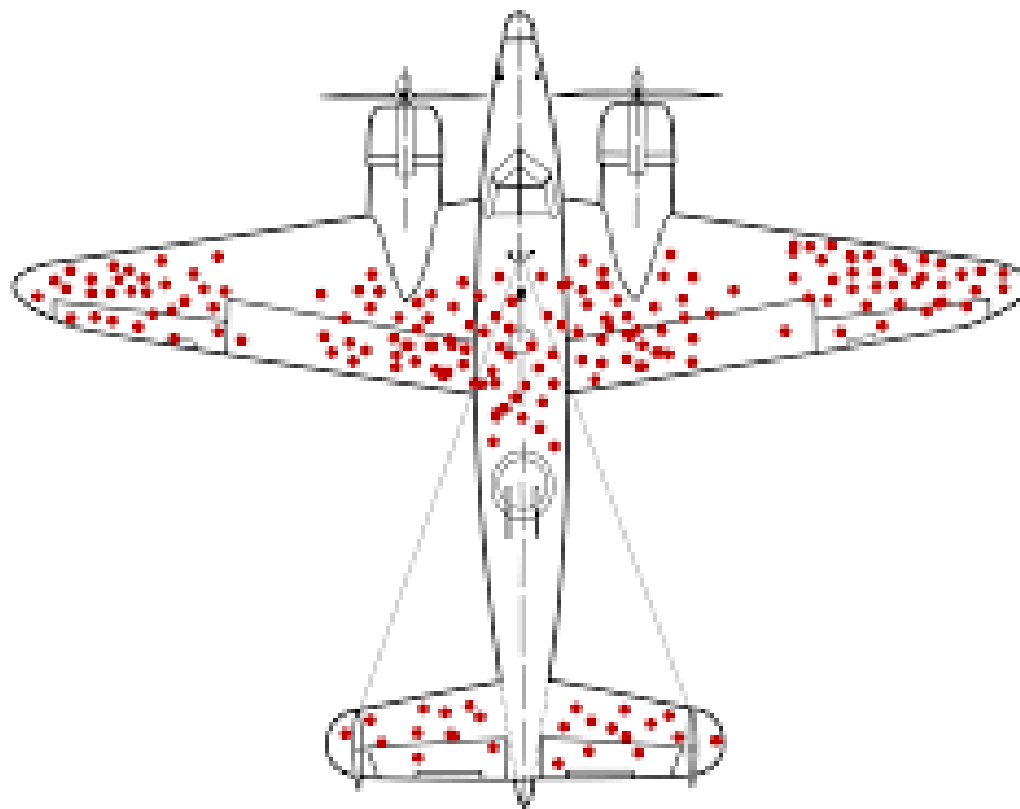


Common: Evaluate the mean difference
In a biomarker in the “Healthy” versus the “Diseased” using Mann-Whitney test.

Failure to conduct and/or report a statistical power analysis

- Power analyses are very important for interpreting null findings (positive findings, too)
 - Without available statistical power, one cannot distinguish between the lack of a 'true' effect versus the lack of power to detect one





All statistical analyses are really exercises in the selection of observations

Missing data (or incompletely observed data)

- Missing data pose a threat to the interpretation of all studies
- Data not collected for all individuals
 - Only those not in acute distress
 - Only those with a certain hospital length of stay
 - Only those animals that survived experimentation
- Biomarker assay not available for all levels
 - Minimal detectable threshold

All Statistical Analyses are really exercises in the selection of observations

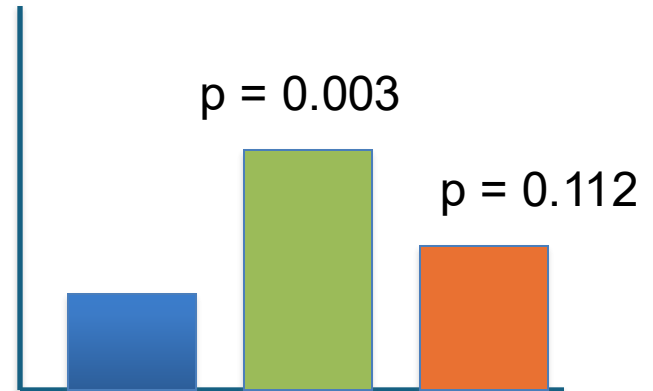
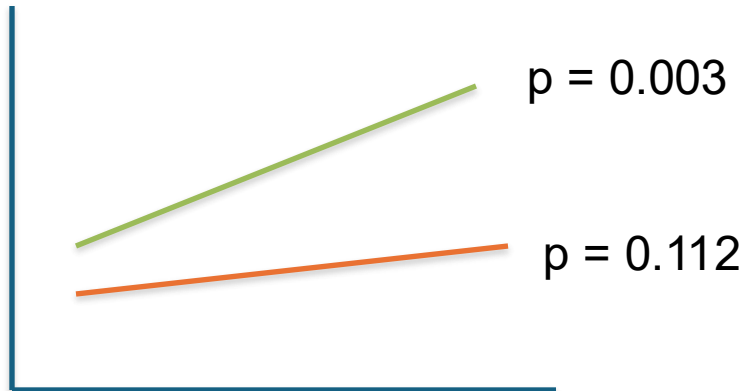
Missing data in submitted manuscripts

What to do about missing data?

- Must know that data are “missing”
 - Reporting of protocol is crucial (e.g., STARD)
 - What patient groups are not being included?
- Characterization of missing data
 - For whom, when, and theoretical effects
 - Standard missing value analysis
- Then, estimation of bias is possible
 - Multiple imputation, clustering methods, etc.
 - Secondary to experimental design (the best way to handle missing data is to not have it)

Comparing p-values as evidence of a differential effect

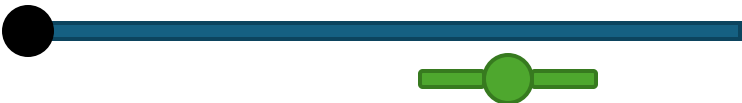
- Only in rare circumstances can p-values be directly compared to conclude differences in groups
- Differences must be directly tested.



The evaluation of a “straw-man” hypothesis test, statistical power issues

“Straw-man” null hypothesis

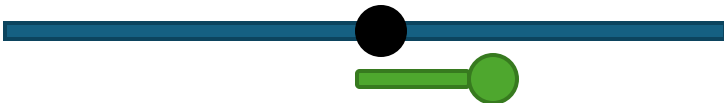
0.78 (95% CI 0.74 to 0.82)



Testing sensitivity/specificity
against zero, $p < 0.0001$

“Interesting” null hypothesis

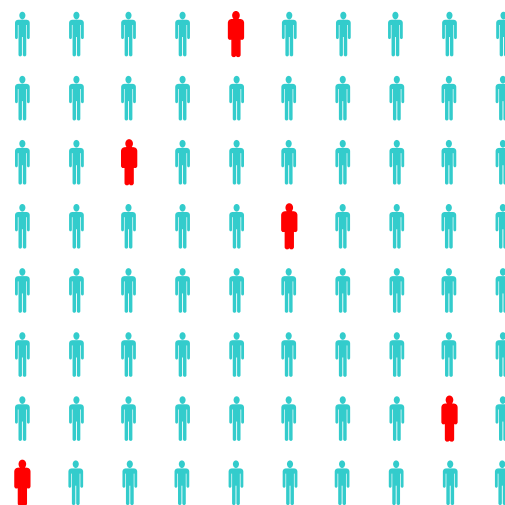
0.78 (95% LB 0.68)



Testing sensitivity/specificity
against some interesting value (0.70), $p = 0.072$

Problem of “over-fitting”, Type-I error rate

- Most biomarker studies (still) rely on statistical inferences for screening
- If screening is used for a pool of biomarker candidates, control for type-I error should be employed
- Several methods could be used, each with their own strengths and weaknesses
 - Bonferroni adjustments
 - False discovery rates (FDR)
 - More all the time



Screening 100 biomarkers
Will likely result in 5 statistically
significant associations ($p < 0.05$)

Statistical significance is a necessary but not sufficient descriptor of the utility of a biomarker

Problem of “over-fitting”, Type-I error rate

- Another problem in the context of statistical models is “over-fitting”
 - Model describes random error instead of underlying relationship
 - The uncovered relationships exploit chance associations in the data
- Occurs when the ratio of predictors (or weights) to outcome is lean, or with multicollinearity (ill-conditioned models)
 - Bias is then expected in the parameter weights
 - The model will not replicate as well as desired
- Can be addressed using a number of simple methods
 - Ample sample size to predictor ratio (e.g., 10:1, 15:1, 30:1)
 - Internal validation techniques (bootstrapping, training/test set)
 - “Phantom” degrees of freedom accounting

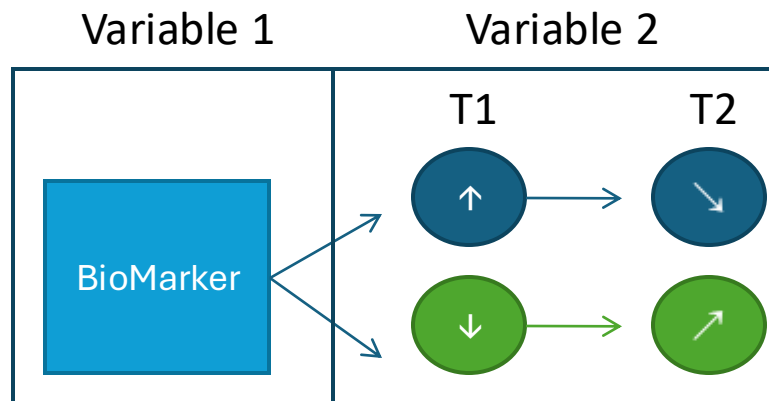
Regression to the mean

- A variable can be used to create “extreme” groups from which to evaluate the effectiveness of treatment using a pre-post test
- Such a design makes regression to the mean a statistical inevitability
 - High scores “get” lower; Low scores “get” higher
 - Occurs whenever there is a non-perfect correlation between the two measurement occasions



Regression to the mean should be considered in biomarker treatment studies

- The preceding example was perhaps overly-simplistic
- However, regression to the mean should be expected for any study involving change
 - For all non-perfectly reliable variables
 - One variable over time OR one variable used to classify a second variable



Dichotomization of a variable that is naturally continuous

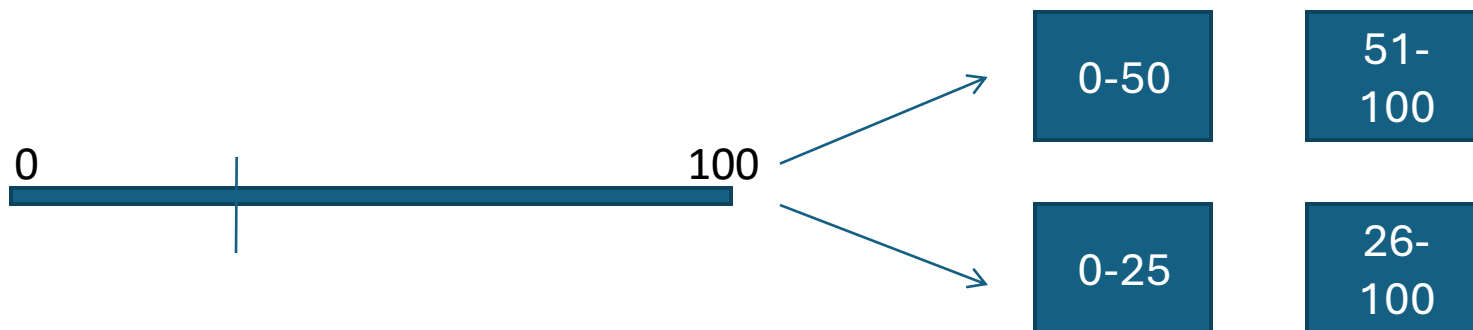
- Although proven to be a poor practice (1993), medical researchers continue to transform continuous variables (e.g., BMI) into discrete categories for analysis (e.g., underweight, normal, obese)
- For clinical purposes this can make sense, as in a decision point
- However for the statistical evaluation of a variable, this choice is quite often a bad idea.



Dichotomization of a variable that is naturally continuous

- Why do this?

- Simplify analysis (yes)
- Simplify interpretation (yes and no)
- Dichotomizing is more reliable than using imprecise measurements (no)
- An “optimal” cut-off can be used (yes, at a cost)



Dichotomization of a variable that is naturally continuous

- Why NOT do this?

- Loss of statistical power
 - When $\sim N(\mu, \sigma)$, analyses are only 65% as efficient
 - Might as well throw away 1/3rd of the data!
- Loss of information
 - Non-linearity cannot be examined
 - Within-group variability absorbed in the entire group
- Scores close to the cut-point are viewed inappropriately
- Dichotomized confounders are not well modeled (residual confounding)
- Ambiguous choice of the “optimal” cut-off

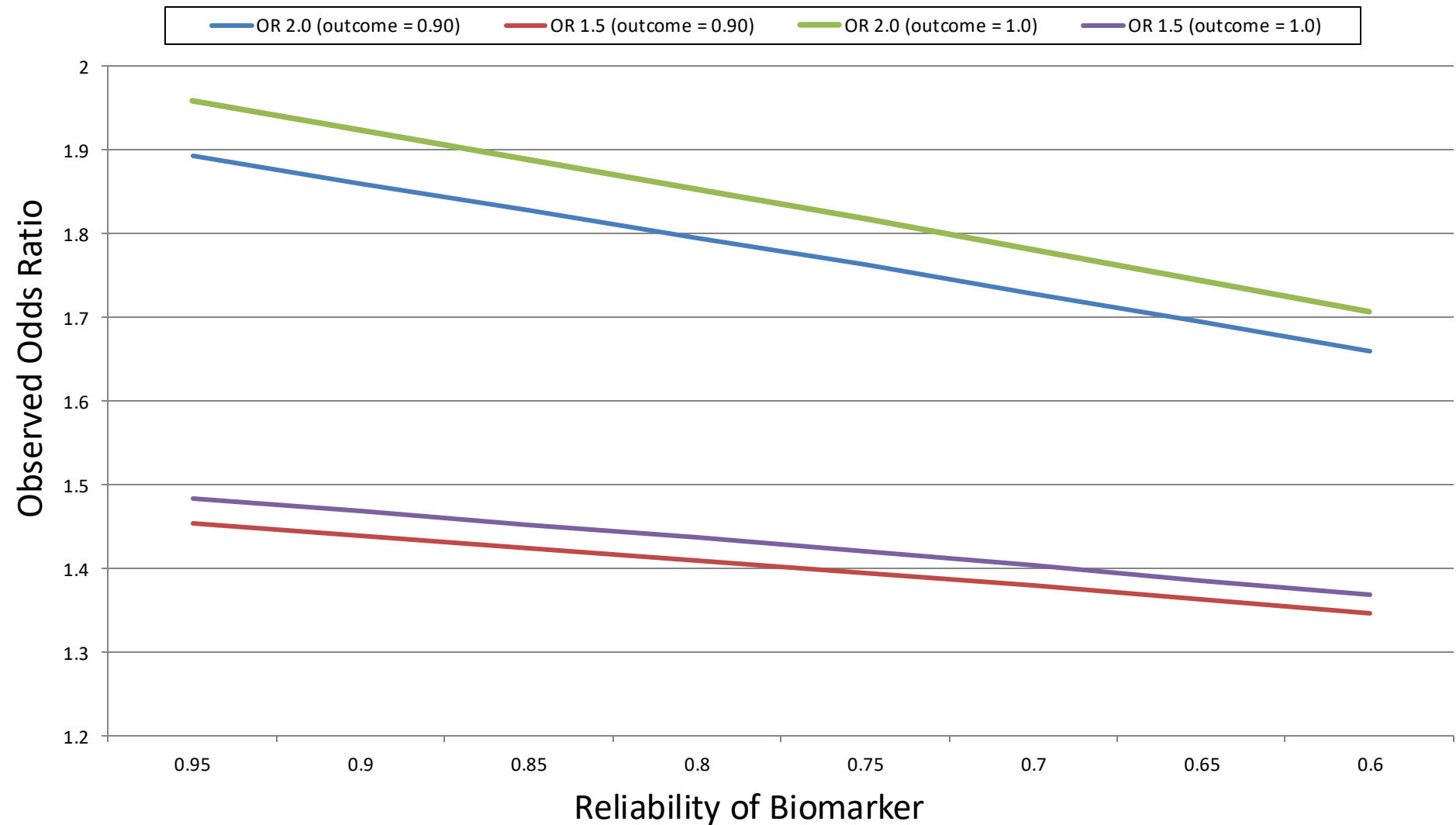
The (un)reliability of measurements

- Studies in medicine seldom explore the statistical reliability of the measurements
 - Perhaps because the variables drawn from a biological sample, it is assumed to have high reliability (i.e., precision).
- However, the measurement of these variables can be obscured by error like any other measurement
 - Measurement error (random error)
 - Imprecision of assay,
 - Systematic error (bias)
 - Diurnal variation of a biologic variable
- Reliability and validity are contextual (i.e., for a specific setting)

$$\text{"True" association}_{xy} = \frac{\text{Observed Association}_{xy}}{\text{Reliability}_x * \text{Reliability}_y} = r_{x'y'} = \frac{r_{xy}}{\sqrt{r_{xx} * r_{yy}}}$$

The reliability of a biomarker/outcome

“True” odds ratio (reliability of outcome)



Thanks!

1. The reliability of a measurement
2. Artificial dichotomization
3. Regression to the mean
4. “Over-fitting”, Type-I error
5. “Straw-man” hypotheses
6. Comparing p-values
7. Missing data
8. Low Statistical Power
9. Assumptions of Tests
10. Poor Reporting